

Assessment 2: Programming Demo of Model Development

10% of final mark, Book your Demo in Week 9, Submit Your Code by Monday, 21/09, 5pm.

This programming demo will comprise an 8-minute presentation of your code, its execution, and will be followed by 4 minutes of reviewer questions. *Submitted code must be identical to your live and recorded demo.* The demo will comprise 100 marks, averaged to be 10% of your final course grade.

At this stage, your programming code should successfully implement and plot simulation results of your formulated model, but should not yet include data processing, analysis, parameterisation, or sensitivity analysis.

Code Presentation You should present your code to your reviewers. This presentation should be separated into the four first laboratory outcomes on the following page's rubric. For instance, outlining the equations that your model uses and how you've programmed them in your model, outlining the constants, initial conditions, scheduling, and writing of these equations, and usage of loops (FOR/IF/WHILE), then presenting how you've programmed the output of your model, for instance using plots. All throughout you should highlight how you've commented your code in these different sections to be clear and understandable to the external user.

Code Execution You will be asked to press "run" on your script on the Anaconda Spyder 3 running environment. You will be receive a score based on whether your code was completely able to run without errors, or if there was errors, how far through the code that simulation could run.

Reviewer Questions You will be asked a series of questions (e.g., 3 questions) pertaining to the above presentation and execution by your reviewers. This will change presentation-to-presentation and will provide an opportunity to describe challenges, future steps, and consideration.

Of potential help, below is the rubric of last years' assessment, which can provide additional guidance:

Model Implementation. [100 points] *The below sections will add up to 100 points together. As these subcategories have a great degree of crossover, grading subcategories separately will be difficult. Points will not be subdivided into categories. Please consider the following items:*

Parameter Set-Up. Program a section of code which assigns all parameters and initial conditions to values, clear and well-commented with appropriate units, consistent with prior parameter table. This section may also comprise some algebraic equations needed to calculate these parameters.

Treatment Scheduling. Program a section of code which schedules treatment to be administered at appropriate, intermittent times across the simulation.

Dynamic Simulation. Program a section of code simulating the differential equations previously described by implementing a FOR loop. This dynamic simulation will have to include treatment scheduling, for instance by implementing IF statements, although other solutions exist.

Plotting & Presentation. Program a section of code which plots one graph for each differential variable (y-axis) over time (x-axis). The simulated time should correspond to the range of time values provided in the *formulation* section. Legends should be programmed into the plots, and a written caption under the plot should be provided in the report.

Error Explanation. While simulations or plotting errors in the above code will reduce final marks, a fraction of these lost marks can be recovered with a explanation of errors that clearly understands the shortcomings made and what should be done to correct them.

Assessment 2: Programming Demonstration Grading Rubric

Laboratory Outcome	Not attempted, or incorrect	Partially complete, or some mistakes/misunderstandings	Complete and correct
Appropriate and sensible conservation and constitutive equations have been used.	0	5	10
The constants, initial conditions, scheduling, equations, FOR or IF or WHILE loops are and consistently laid out	0	10	20
Results and figures are sensibly plot and formatted (with correct titles, axes, and legends)	0	10	20
Code is clearly formatted and commented to an appropriate detail	0	5	10
Code Execution: Code can run	0	10	20
Reviewer Questions are answered correctly and confidently	0	10	20
TOTAL MARKS	/100 points		